Radiation - Student Notes

Part 1: Introduction to Radiation | Part 2: Basic Terms and Black Body

Part 1 - Introduction to Radiation

Radiation is the transfer of energy through electromagnetic waves.

Unlike conduction and convection, radiation does not require a material medium. It can travel through vacuum.

1. Three basic facts

• Every body above 0 K emits radiation continuously.

• Radiation needs no medium. It can travel through vacuum at the speed of light.

• A body at the same temperature as its surroundings still radiates. It absorbs radiation from the surroundings at the same rate, so there is no net heat loss.

If a body and its surroundings are at the same temperature:

Energy emitted = Energy absorbed => Net heat transfer = 0

Important: No net heat transfer does not mean radiation has stopped. Emission and absorption continue simultaneously.

Part 2 - Absorption, Emission and Black Body

1. What happens when radiation falls on a surface?

The incident radiation can be absorbed, reflected or transmitted.

$$a + r + t = 1$$

Symbol	Meaning
a	absorptive power - fraction of incident radiation absorbed
r	reflectance - fraction of incident radiation reflected
t	transmittance - fraction of incident radiation transmitted

2. Absorptive power, a

Absorptive power is the fraction of incident radiation absorbed by the surface.

$$a = \text{Energy absorbed} / \text{Energy incident and } 0 <= a <= 1$$

Example: a = 0.8 means the surface absorbs 80% of the radiation falling on it.

3. Emissive power, E

Emissive power is the energy radiated by a body per unit area per unit time.

$$E = \text{Energy radiated} / (\text{Area} \times \text{Time})$$

SI unit: W/m². Emissive power depends strongly on temperature.

4. Emissivity, e

Emissivity tells how good a surface is at emitting radiation compared with a perfect emitter at the same temperature.

$$0 <= e <= 1$$

It is a pure number and has no unit. If e = 0.6, the surface emits 60% of what a perfect emitter would emit at the same temperature.

Emissive power and emissivity are different

Emissive power, E	Emissivity, e
Actual radiation emitted per unit area per second	Quality of a surface as an emitter compared with a perfect emitter
Unit: W/m²	No unit; value lies between 0 and 1

5. Perfectly black body

A perfectly black body absorbs all radiation falling on it.

$$a = 1, r = 0, t = 0, \text{ and emissivity } e = 1$$

6. Practical black body - cavity with a small hole

A hollow cavity with a small hole behaves approximately like a black body. Radiation entering the hole undergoes repeated reflections inside the cavity. At each reflection, part of the radiation is absorbed, so almost none escapes back through the hole.

> The small hole behaves as the black body, not the entire box.

The pupil of the eye and a dark window viewed from outside are examples of openings that appear black because very little light returns from inside.

7. Important note

> Black body does not mean black-coloured body. The term refers to perfect absorption. At sufficiently high temperature, a black body can glow red, yellow or white.